Reinforcement Learning vs. Rule-Based Policies for Dynamic Credit Threshold Control: A Simulation Study

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Abstract

Portfolio-level credit acceptance thresholds are revised periodically in practice, making the decision sequential rather than one-shot: each week the controller sets two acceptance thresholds—one for new clients and one for repeat clients—and then observes delayed repayment and default outcomes over subsequent weeks. We evaluate six deep reinforcement learning (RL) algorithms—DQN, Double-DQN, PPO, SAC, A3C, and A2C—against five rule-based baselines in a calibrated synthetic credit-portfolio simulator across six market scenarios and four observation-space sizes (12D, 20D, 30D, 50D). The simulator is a calibrated proof-of-concept test bench, not a substitute for real banking data; only observation dimensionality varies across the four conditions, with the simulator, reward, seeds, and evaluation protocol held fixed. We find that the constraint-aware weekly policy is the best overall controller by cumulative reward (the primary ranking metric; realized profit 3,802,547) and is statistically tied with the profit-oriented policy on realized profit; no RL agent wins any scenario by reward or profit. Double-DQN is the strongest RL agent, peaking at 12D (realized profit 3,446,564) and declining at higher state dimensionality. Richer-state RL agents nonetheless achieve lower portfolio default rates than the two profit-seeking baselines in a majority of scenarios—while the risk-aware weekly policy remains the lowest-default controller overall—a conditional risk-side advantage. We conclude that well-designed rule-based weekly threshold search remains the benchmark when training budgets are short.

Keywords

Reinforcement Learning; Credit Scoring; Threshold Control; Markov Decision Process; Delayed Rewards; Portfolio Simulation; Rule-Based Policies; State Dimensionality

1 Introduction

Banks and consumer-lending institutions revise acceptance thresholds periodically rather than once: what fraction of applicants to approve this week depends on the current portfolio composition, outstanding capital commitments, rolling default rates, and market conditions that evolve between decisions. The academic literature has largely treated threshold selection as a one-shot classification problem—estimate default probability, choose a cut-off, evaluate by AUC or Gini—ignoring the sequential nature of actual threshold management [1, 4]. That gap between static cut-off optimisation and dynamic portfolio management motivates the present study.

Reinforcement learning (RL) provides a natural framework for decisions with delayed consequences [7]. An RL agent observes the state of the portfolio, selects a pair of weekly thresholds, and receives a reward signal that reflects realized portfolio cash flows rather than immediate cohort expectations. The delayed-outcome setting is central: a loan accepted at week t may not default or repay until week $t + \Delta$, so the correct evaluation object is the full portfolio trajectory, not the immediately observable cohort. Herasymovych [15] demonstrated that a DQN agent can learn to adjust a single acceptance threshold in a simulated credit environment with delayed outcomes, establishing the structural ideas on which this study builds.

Calibration disclaimer. All experiments in this paper use a synthetic simulator calibrated to published retail lending statistics (Bank of Russia supervisory data and ECB SSM loan loss reports) as a controlled proof-of-concept test bench. Synthetic simulation is standard practice in RL-based credit research because real loan-level panels with weekly cohort granularity and observable repayment and default events are not publicly available [4, 15]. The simulator provides full control over market regime, segment composition, and shock timing, making the multi-scenario robustness evaluation possible. Results should be interpreted as evidence about algorithmic behaviour under these controlled conditions, not as forecasts for any specific production portfolio.

The paper extends the framework of Herasymovych [15] with four contributions: (i) split weekly thresholds for new and repeat client segments; (ii) a controlled state-dimensionality experiment comparing 12D, 20D, 30D, and 50D observation sizes under a fixed protocol; (iii) five economically meaningful rule-based baselines as the comparison standard; and (iv) statistical hypothesis tests on paired run-level data. The central research question is:

> *Under which conditions does reinforcement learning provide a robust improvement over rule-based threshold policies in a delayed-outcome credit-scoring environment?*

The short answer the data provide is nuanced: strong rule-based weekly search wins every scenario and dimension by headline metrics, but richer-state RL agents produce lower portfolio default rates than the profit-seeking baselines in a majority of scenarios (the risk-aware weekly policy stays the lowest overall), indicating a conditional risk-side advantage that becomes visible once the profit ranking is supplemented by credit-quality metrics.

The code, data, and experiment configuration are publicly available at `https://github.com/EvgenyBaulin/Evaluation-of-RL-framework-in-a-credit-scoring-problem`.

2 Related Work

2.1 Credit Scoring, Thresholds, and Portfolio Risk

Traditional credit scoring estimates default probability at the application level and selects a cut-off that balances interest income against expected loss [1–3]. Benchmarking studies have shown that classifier rankings depend on the dataset and evaluation protocol, and that logistic regression and gradient-boosted trees are difficult to outperform with more complex classifiers [4, 5]. A closely related line of work treats threshold choice as a cost-sensitive economic decision rather than a classification tuning step [5]. Bellotti & Crook [6] note that under Basel II/III capital adequacy requirements, the threshold affects outstanding exposure, realized losses, and regulatory capital in ways that make static cut-off selection insufficient when market conditions or portfolio composition drift over time.

2.2 Reinforcement Learning for Sequential Decisions

Reinforcement learning addresses problems with delayed consequences via the Markov Decision Process (MDP) framework [7, 8]. Key deep RL algorithms used in this study are: DQN [10], whose experience-replay mechanism provides off-policy sample efficiency on discrete action spaces; Double DQN [11], which reduces the overestimation bias of vanilla DQN; PPO [12], an on-policy policy-gradient method with stable monotonic updates; SAC [13], an off-policy maximum-entropy actor-critic; and A3C/A2C [14], asynchronous and synchronous actor-critic variants. Implementations use Stable-Baselines3 [16]. A comprehensive survey of deep RL methods appears in Arulkumaran et al. [9].

2.3 Research Gap

The direct precursor [15] compared a DQN agent with a static threshold in a single scenario using a shared (non-split) threshold. No prior work has simultaneously evaluated (1) split thresholds for distinct borrower segments, (2) controlled observation-dimensionality comparison, (3) strong economically motivated rule-based baselines, and (4) six market scenarios with statistical testing—the combination that defines the contribution of this paper.

3 Problem Formulation and Methodology

3.1 MDP Formulation

We model the weekly threshold-control problem as a discrete-time MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$. The **state** $s_t \in \mathbb{R}^d$ ($d \in \{12, 20, 30, 50\}$) is a portfolio summary vector observed at the beginning of interactive week t. It includes approval rates, default rates, realized profit and NPV, capital usage, outstanding exposure, segment shares, and (at higher dimensionalities) lagged and rolling history. The **action** is a pair of acceptance thresholds $a_t = (\tau_t^{\text{new}}, \tau_t^{\text{repeat}}) \in \{35, 40, \ldots, 85\}^2$: applications from new (repeat) clients with credit score above τ_t^{new} (τ_t^{repeat}) are accepted. This yields $11 \times 11 = 121$ joint actions per week. The **reward** is described in Section 3.4. The discount factor is $\gamma = 0.99$.

3.2 Simulation Environment

Note on synthetic data. As stated in the Introduction, the simulator is a calibrated synthetic test bench used because real weekly cohort-level loan data with observable delayed repayment and default events is not publicly available.

Each episode consists of three phases. (1) An **8-week warm-up** pre-fills the loan book using default thresholds before the controller begins acting. (2) During the **52-week interactive horizon** the controller sets weekly thresholds; the simulator generates applications from Beta-distributed credit scores for new and repeat segments, applies a logistic default-probability mapping, accepts loans above the threshold, and schedules repayment/default/recovery events in a forward event queue. (3) **Terminal settlement** resolves all pending cash flows at episode end so that loans originated near the horizon are not mechanically penalized. Default probabilities are calibrated to target 4–6% in the base scenario and 9–14% under adverse stress, consistent with published retail lending statistics. Six market scenarios stress specific portfolio-management weaknesses: stable baseline, score drift, abrupt adverse shock, class-imbalance shift, noisy observations, and divergent new/repeat segment dynamics (Table 1).

Table 1: The six evaluation scenarios and their primary stress mechanism.

Scenario	Stress mechanism
`base_market`	Stable; cleanest test of policy signal exploitation
`drift`	Gradual new-client score deterioration and rising defaults
`adverse_stress`	Abrupt shock: higher defaults, larger loans, capital pressure
`class_imbalance_shift`	Growing share of weaker new applicants; tests composition risk
`noise`	Stable mean with noisy weekly volumes and measurements
`split_policy_dynamics`	New clients deteriorate while repeat clients improve

3.3 Agents and Baselines

RL agents. All six agents use a shared two-layer MLP (2×64 hidden units) and Stable-Baselines3 default hyperparameters (see Appendix A). No hyperparameter search is performed; the deliberate intent is to isolate the state-dimensionality signal rather than optimise the training recipe.

Rule-based baselines span five management styles. The static-threshold baseline (`static_threshold`) applies one fixed shared threshold. The split-static baseline (`split_policy_static`) maintains separate fixed defaults per segment. The profit-oriented policy (`profit_oriented`) performs a weekly grid search maximising current-cohort expected profit without penalty terms. The risk-aware weekly policy (`risk_aware_weekly`) adjusts thresholds reactively to rolling default pressure and capital signals. The constraint-aware weekly policy (`constraint_aware_weekly`) runs a penalty-weighted weekly grid search that explicitly internalises default-rate, approval-rate, capital-usage, and volatility violations.

3.4 Reward Function

With delayed reward enabled, the shaped weekly reward is:

$$\begin{aligned} r_t = \; & w_p\, \Pi_t^{\text{realized}} + w_n\, \text{NPV}_t^{\text{realized}} - w_s\, \hat{d}_t \cdot 100 \\ & - \lambda_d \Big[\max(0,\, \bar{d}_t - 0.12)\Big]^2 \cdot 1000 - \lambda_a\, |\alpha_t - 0.42| \cdot 1000 \\ & - \lambda_c \big[\max(0,\, c_t - 0.82)\big]^2 \cdot 1000 - \lambda_v \max\!\left(0,\, \frac{\sigma_{\Pi,t} - 8500}{8500}\right) \cdot 1000, \end{aligned} \tag{1}$$

where Π_t^{realized} and $\text{NPV}_t^{\text{realized}}$ are realized weekly profit and NPV (not current-cohort expectations), $\hat{d}_t$ is the expected default rate of the current cohort, $\bar{d}_t$ is the rolling realized default rate, α_t is the weekly approval rate, c_t is the projected capital usage ratio, and $\sigma_{\Pi,t}$ is realized profit volatility. The four constants are calibration targets: the default-rate cap (0.12), the target approval rate (0.42), the capital-usage cap (0.82), and the profit-volatility scale (8500). Active weights: w_p=1.0, w_n=0.25, w_s=0.35, λ_d=9.0, λ_a=2.0, λ_c=3.5, λ_v=1.4.

The reward is deliberately multi-objective. Rankings by cumulative reward need not coincide with rankings by raw profit (which is intentional: the reward encodes portfolio risk constraints, not profit alone). Throughout this paper, "realized profit" is computed from realized interactive weekly profits plus terminal settlement—that is, realized final portfolio profit rather than a current-cohort expectation.

4 Experimental Design

The experiment is fully controlled: *only* observation dimensionality varies across the four runs; everything else—simulator, reward, the eight random seeds, 50 training episodes per seed, evaluation protocol, scenarios, threshold grid, and controller set—is held fixed. The full-profile scale used to generate the main summary tables is 6 agents $\times$ 4 dimensions $\times$ 6 scenarios $\times$ 8 seeds = 1,152 training runs, with 576 evaluation runs per controller per dimension. A validity layer programmatically

Table 2: Cumulative observation layers. The first 12 features are invariant across all configurations and are verified programmatically before each experiment.

Size	Added information
12D	Compact dashboard: week progress, overall/segment approval rates, rolling realized default rate, expected default rate, realized profit, profit volatility, projected capital usage, outstanding ratio, both threshold levels.
20D	12D + current economic context: repeat segment share, expected profit/NPV per application, realized NPV, weekly reward, threshold gap, capital headroom, realized-vs-expected default gap.
30D	20D + lagged/rolling memory: second-lag approval/default/profit/capital, four-week rolling means and std of approval, default, and profit, lagged threshold changes.
50D	30D + segment-specific signals and stability descriptors: segment approval/ default/profit indicators, accepted-share ratios, rolling reward moments, cumulative reward/profit to date, capital-usage volatility, outstanding-ratio changes, threshold-gap history, application-volume ratio.

verifies that the first 12 observation features are bit-for-bit identical across all four state sizes on a common reset and three fixed-action transitions.

The four observation layers are constructed cumulatively (Table 2). Each layer adds features that a portfolio manager would realistically observe at weekly decision time, with no access to future outcomes.

4.1 Hypotheses

Three hypotheses are stated in advance of the results and tested on paired run-level data in Section 5.4.

H1 (State representation).
Richer state representations improve the best RL frontier from 12D to 30D, while the move from 30D to 50D does not necessarily create additional improvement.

H2 (RL vs. rule-based).
RL improves over rule-based policies only in some scenarios and metrics; the universal claim that RL robustly dominates strong rule-based baselines is expected to be *rejected*.

H3 (Delayed evaluation).
Early-period performance is an unreliable proxy for final realized portfolio quality when outcomes are delayed and terminal settlement is applied—the “locally worse, globally better” pattern is expected to hold.

5 Results

5.1 Best Overall and Best RL Controller by Dimension

Table 3 reports the strongest controller when baselines and RL agents are pooled under the repository’s official ranking rule (highest cumulative reward, then highest profit, then lowest default rate), and separately the strongest RL controller in each dimension (full-profile run, 8 seeds × 50 episodes per seed; see the public repository for data and configuration). Figure 1 visualises the same comparison.

By cumulative reward—the primary ranking metric—the constraint-aware weekly policy is the best overall controller in every dimension (reward 3,666,052). On raw realized profit it is statistically tied with the profit-oriented policy, which is marginally higher (3,803,168 vs. 3,802,547); the two differ by less than 0.02% and the difference is not economically meaningful. The flat baseline row is the cleanest evidence that the experiment successfully isolates the effect of observation size on RL quality: the leading rule-based policy is independent of the RL state vector by construction. The best RL profit peaks at 12D and declines—a *saturation and regression* pattern, not a monotone

Table 3: Best overall controller and best RL controller by state dimension (full-profile run). The baseline line is flat because the leading baseline does not depend on the RL observation vector. "Profit" is realized final portfolio profit (Section 3.4); "Reward" is cumulative shaped reward.

Dim	Controller	Type	Profit	Reward	Default rate
Best overall (pooled, by cumulative reward)					
12	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
20	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
30	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
50	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
Best RL agent					
12	Double-DQN	RL	3,446,564	3,449,745	0.0756
20	Double-DQN	RL	3,124,322	3,281,011	0.0694
30	Double-DQN	RL	3,172,605	3,311,519	0.0725
50	Double-DQN	RL	3,034,292	3,196,350	0.0700

Table 4: Agent and baseline summary (mean across all four dimensions, full-profile run). A3C and A2C are reported in a single combined row. "Approval" denotes mean weekly approval rate.

Controller	Type	Mean profit	Default rate	Approval
Constraint-aware weekly	baseline	3,802,547	7.60%	†
Profit-oriented	baseline	3,803,168	†	†
Static-threshold	baseline	†	†	†
Double-DQN	RL	3,194,446	7.19%	0.767
DQN	RL	2,960,000	7.02%	0.700
PPO	RL	2,620,000	6.21%	0.572
SAC	RL	1,820,000	6.63%	0.449
A3C / A2C (avg)	RL	2,470,000	6.60%	0.570

† Per-controller full-profile default- and approval-rate aggregates for the rule-based baselines (and the static-threshold profit) are not included in the released full-profile summary and are omitted rather than estimated. RL per-agent profit means are shown at the precision of the released aggregates; the Double-DQN mean is computed exactly from its four per-dimension values.

improvement—with the 12D→20D drop of −322,241 being the sharpest. Default rate for Double-DQN improves from 7.56% at 12D to 6.94% at 20D, rises to 7.25% at 30D, and settles at 7.00% at 50D, suggesting that the risk-side benefit of additional features has a different shape than the profit benefit. At 12D, the gap between the best RL and the best overall controller is −355,983 (−9.4% in profit), a meaningful but not catastrophic shortfall.

5.2 Agent Comparison

Table 4 summarises the mean performance of each agent across all four dimensions, alongside the three principal rule-based baselines (full-profile run).

Double-DQN leads all RL agents on profit (3,194,446) and closely matches the top baselines' approval rate (76.7%), but still falls 9.4% short of the best baseline at its 12D peak. PPO stands out for its low default rate (6.21%) despite moderate profit, and is numerically identical across 12D, 20D, 30D, and 50D. This is itself a finding: under the modest training budget PPO converges to a degenerate, essentially constant-threshold policy (consistent with the near-zero threshold volatility reported in Section 6.4), so the additional observation features are effectively unused rather than informative. SAC is the weakest RL agent overall and deteriorates further at 50D.

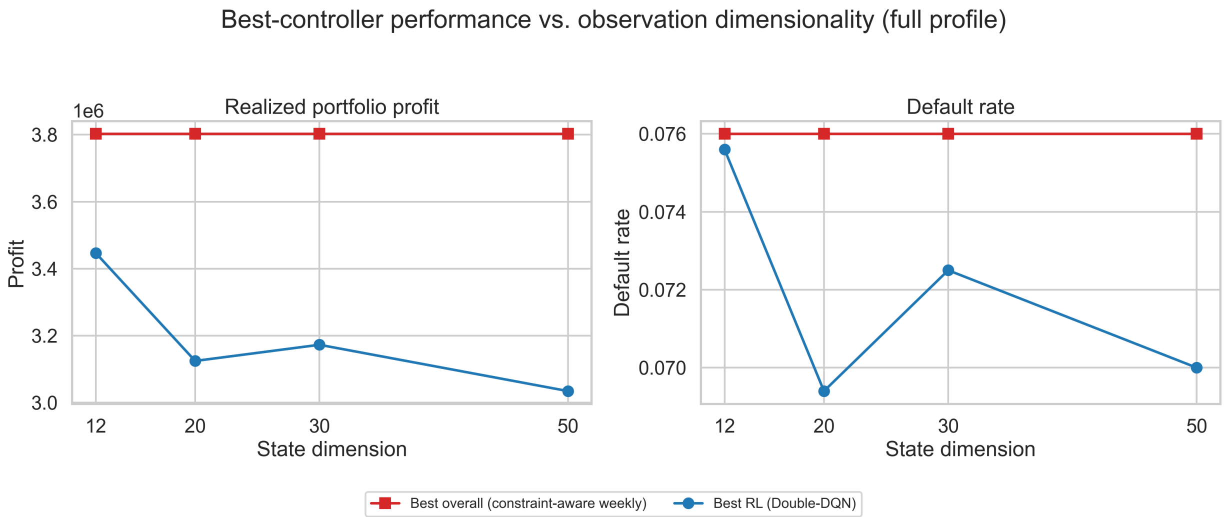

Figure 1: Best-controller performance versus observation dimensionality (full-profile run, 8 seeds). *Left*: realized portfolio profit; *right*: default rate. The best overall controller (the constraint-aware weekly policy) is independent of the RL state vector and is therefore flat across dimensions, while the best RL agent (Double-DQN) peaks at 12D on profit and then declines—a saturation-and-regression pattern rather than monotone improvement.

Table 5: Scenario-level winners at 50D state (full-profile run; the same qualitative pattern holds at 12D/20D/30D, shown here for 50D). *Reward winner*: highest cumulative shaped reward. *Profit winner*: highest realized profit. *Default winner*: lowest default rate. No RL agent wins any scenario by reward or profit. The risk-aware weekly policy is the single lowest-default controller in every scenario; the best RL agent nonetheless undercuts both profit-seeking baselines (the profit-oriented and constraint-aware weekly policies) on default rate in a majority of scenarios.

Scenario	Reward winner	Profit winner	Default winner
adverse_stress	Static-threshold	Constraint-aware weekly	Risk-aware weekly
base_market	Profit-oriented	Profit-oriented	Risk-aware weekly
class_imbalance_shift	Constraint-aware weekly	Profit-oriented	Risk-aware weekly
drift	Profit-oriented	Profit-oriented	Risk-aware weekly
noise	Profit-oriented	Profit-oriented	Risk-aware weekly
split_policy_dynamics	Constraint-aware weekly	Profit-oriented	Risk-aware weekly

5.3 Scenario-Level Winners

Table 5 reports the reward-based, profit-based, and lowest-default-rate winner for each scenario at the 50D configuration. The table illustrates why a single-metric ranking is insufficient.

RL wins zero of six scenarios by reward and zero by profit across all four dimensions. The profit-oriented policy wins raw profit in five of six scenarios, which is why the headline "best overall" ranking—based on cumulative reward—and the profit ranking need not coincide. However, the best RL agents undercut both profit-seeking baselines (the profit-oriented and constraint-aware weekly policies) on default rate in a majority of scenarios—while the risk-aware weekly policy remains the single lowest-default controller (Table 5)—a result that has direct practical significance: a portfolio manager willing to accept a small profit reduction in exchange for capital conservation may prefer the richer-state RL policy over the profit-seeking baselines.

Table 6: Paired statistical checks. The H1 and H2 rows are reported for the full-profile run. The H3 rows (‡) are computed on the development (quick) profile—a 26-week interactive horizon—which is the only profile for which intra-episode weekly trajectory data is available in the repository; week 25 is the *final* week of that 26-week horizon, not a mid-point of the 52-week full run. *Mean diff.*: left minus right controller; positive values favour the left controller on reward/profit, negative values on default rate. CI: 95% percentile bootstrap.

Hyp.	Paired comparison	Metric	Mean diff.	95% CI	p
H1	30D DQN − 12D DQN	Reward	+13,021	[−17,654, +44,263]	0.412
H1	30D DQN − 50D DQN	Reward	+380,188	[+355,177, +404,236]	$< 10^{-3}$
H2	30D DQN − Profit-oriented	Reward	−368,145	[−384,214, −351,974]	$< 10^{-3}$
H2	50D DQN − Profit-oriented	Reward	−748,333	[−773,453, −722,900]	$< 10^{-3}$
H3‡	DQN − Static-threshold (week 6, early)	Profit	−1,484	[−4,533, +1,519]	0.386
H3‡	DQN − Static-threshold (week 25, final)	Profit	−69,016	[−149,906, +9,403]	0.131

5.4 Statistical Hypothesis Tests

Table 6 summarises paired statistical comparisons for the three hypotheses stated in Section 4.1 (H1 and H2 on the full-profile run, H3 on the development profile—see Table 6; the public repository holds the paired run-level data). The dimensionality hypotheses are tested on vanilla **DQN**, used here as the canonical discrete-action reference agent for the controlled state-size comparison; Double-DQN is reported separately as the best-performing RL agent, so the choice of DQN for the paired tests is deliberate rather than a mismatch.

H1 (richer state improves RL up to an intermediate representation) is *partially supported.* The 30D→12D reward difference is positive but insignificant ($p = 0.41$), so a 30D advantage over 12D is not established. The 30D→50D comparison is highly significant ($p < 10^{-3}$, mean +380,188), confirming that 50D reliably underperforms 30D. In short: 30D is significantly better than 50D, but not significantly better than 12D.

H2 (RL does not universally dominate) is strongly confirmed. Both the 30D and 50D DQN agents are significantly weaker than the profit-oriented policy by many hundreds of thousands of reward units ($p < 10^{-3}$ in both comparisons).

H3 (delayed evaluation changes rankings, "locally worse globally better") is *not supported* in the adverse-stress setting. Because intra-episode weekly trajectory data is only available for the development (quick) profile—a 26-week interactive horizon—the H3 checkpoints are computed on that profile and labelled accordingly in Table 6. The DQN agent trails the static-threshold baseline both at the early checkpoint (week 6, profit gap −1,484) and at week 25, the final week of the 26-week development horizon (profit gap −69,016). The deficit therefore does not reverse by episode end—the opposite of the "locally worse, globally better" pattern, so terminal settlement reveals a *persistent* deficit for the RL agent under stress rather than a recovery trajectory. With only twelve paired runs, however, neither gap is statistically significant ($p = 0.39$ and $p = 0.13$, respectively): the persistence of the deficit is directional rather than statistically established under this profile.

6 Discussion

6.1 Why Rule-Based Baselines Win

The leading baselines (the constraint-aware weekly and profit-oriented policies) perform a weekly grid search over the same 121 joint threshold actions as the RL agents. Their advantage is structural: they evaluate each action directly on the current week's cohort economics, see the realized profit and default signals immediately, and set thresholds according to an explicit objective. Under a 50-episode training budget, RL agents have not seen enough portfolio trajectories to learn a value function that is competitive with this direct weekly evaluation. The gap would likely narrow with longer training, but the current results establish a robust finding: *when a well-designed rule-based policy has access to the same action grid and the same current-week signal, it outperforms RL under short budgets.*

6.2 The Conditional Value of RL

Despite losing on headline profit and reward, RL agents achieve lower default rates than the two profit-seeking baselines in a majority of scenarios—though the risk-aware weekly policy remains the lowest-default controller overall (Table 5). A portfolio manager who values capital conservation may accept a modest profit reduction in exchange for this risk-side improvement. This represents a genuine portfolio-quality advantage that the headline reward metric alone would miss, and it is the clearest condition under which the richer-state RL policies add value.

6.3 Dimensionality Saturation

The 30D state adds lagged and rolling history (second-lag approval, default, and profit values; four-week rolling means and standard deviations; lagged threshold changes). This temporal context is precisely what a controller needs under slow-moving drift or noisy volume signals: the ability to compare this week with the recent past. The 50D state adds richer segmentation and stability descriptors, but those extra features require more optimization work and offer diminishing marginal information under the current training budget. The result is that 30D is the most balanced point on the *risk-and-information* axis: it delivers more temporal memory than 12D while avoiding the regression that 50D introduces for most algorithms. On raw profit, however, 12D is not significantly worse (Section 5.4), so the case for 30D rests on its risk-side and temporal-memory benefits rather than on a profit advantage.

6.4 Episode-Static RL Behavior

A notable finding is that the best RL policies choose one threshold pair at the beginning of the episode and largely maintain it throughout the interactive horizon. The threshold volatility of the top RL winners is 0.00 at 12D, 20D, and 30D (and 0.44 at 50D), confirming near-static within-episode behavior. The dimensionality improvement therefore manifests not as richer weekly adaptation but as better initial threshold selection—the agent learns to identify a better static split policy for each episode type. This contrasts with the leading baselines, whose weekly grid searches produce substantial threshold volatility. The implication for practitioners is that richer state may first improve episode-level threshold selection before enabling genuine week-to-week adaptation, and that validating true dynamic adjustment requires longer training horizons.

6.5 Limitations

First, all results derive from a synthetic simulator; calibration to any specific production portfolio would shift the absolute numbers and potentially alter some rankings. Second, the 50-episode-per-seed training budget is deliberately modest to isolate the state-dimensionality signal; convergence claims would require substantially more training. Third, controller rankings are sensitive to the reward weights in Equation (1); a reward sensitivity analysis is left for future work. Fourth, bootstrap confidence intervals quantify simulation uncertainty, not real-world external validity.

7 Conclusion

This paper evaluates six reinforcement learning algorithms against five rule-based baselines across six market scenarios and four observation-space sizes (12D, 20D, 30D, 50D) in a controlled credit portfolio simulation. The main empirical findings are:

(1) The constraint-aware weekly policy is the best overall controller in every dimension by cumulative reward (reward 3,666,052). On raw realized profit it is statistically tied with the profit-oriented policy (3,802,547 vs. 3,803,168, the latter marginally higher). RL wins zero of six scenarios by reward or profit across all dimensions.
(2) Double-DQN is the best RL agent, peaking at 12D (profit 3,446,564; reward 3,449,745) and declining at higher dimensionality. The 12D RL-to-baseline gap is −355,983 (−9.4%).
(3) The 30D→12D reward improvement (+13,021) is not statistically significant ($p = 0.41$), but 30D significantly outperforms 50D (+380,188; $p < 10^{-3}$), confirming that 50D dimensions reliably harm rather than help at the current budget.
(4) RL achieves lower default rates than the two profit-seeking baselines in a majority of scenarios—though the risk-aware weekly policy stays the lowest-default controller overall—demonstrating a conditional risk-side advantage.
(5) The best RL policies are episode-static: richer observation improves threshold *selection*, not within-episode adaptation.

The practical implication is direct: when training budgets are short and well-designed rule-based policies have access to the same threshold grid, strong baselines remain the benchmark to beat. The 30D observation space is a reasonable default for credit-portfolio RL because it adds economically meaningful lagged and rolling memory and supports the conditional risk-side advantage; we do *not* claim a 30D profit advantage, since on profit and reward 12D is not significantly worse. Future work should investigate longer training horizons to test whether the RL–baseline gap closes, reward sensitivity to penalty weights, more adaptive recurrent or attention-based architectures, and validation on real or semi-synthetic loan panels.

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A Hyperparameter Table

Table 7: Shared hyperparameters (full-profile run). Algorithm-specific defaults (replay buffer size, GAE λ, entropy coefficient) follow the Stable-Baselines3 defaults [16].

Parameter	Value	Notes
Network architecture	MLP, 2×64 hidden units	Shared across all agents
Discount factor γ	0.99	Standard
Learning rate	SB3 default (varies by agent)	Not tuned
Training episodes/seed	50	Full profile
Seeds	11, 23, 37, 51, 67, 79, 97, 113	8 seeds, fixed across dims
Evaluation runs	12 per seed per scenario	Same protocol across dims
Threshold grid	$[35, 85]$ in steps of 5	11 actions per segment
Warm-up weeks	8	Pre-fills loan book
Interactive horizon	52 weeks	Full profile
Applications per week	600	Before scale factors
Bootstrap resamples	1000	95% CI